Appendix

AI use declaration

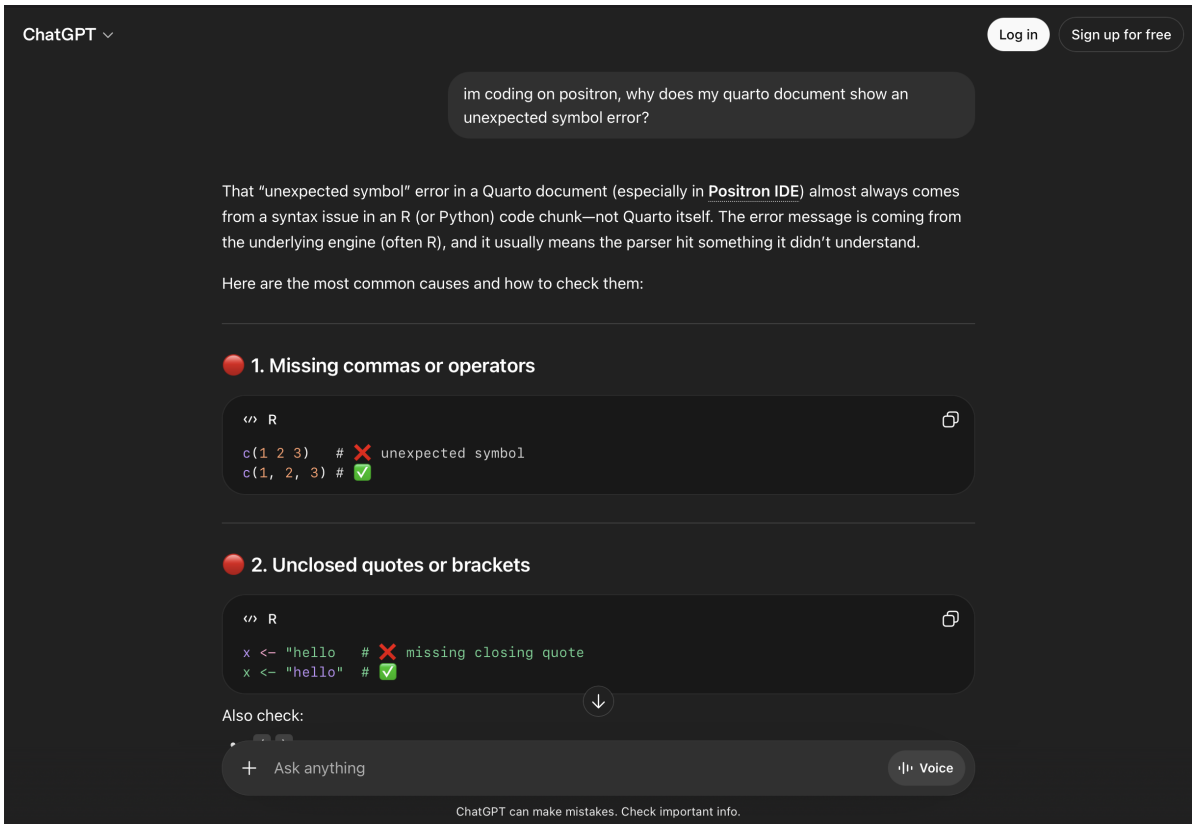
I used generative AI to help debug and improve my Quarto and R code for this assignment. The main analysis, interpretation, and final report wording were my own work. AI was used as a support tool for troubleshooting, formatting, and minor enhancements.

AI queries and prompts

I used AI for the following assignment related questions and code fixes:

"im coding on positron, why does my quarto document show an unexpected symbol error?" "how do I add a table of content to my quarto doc and number my sections" "when im trying to render my quarto doc, it updates the html file ive asked it to create but not the pdf file. this is after ive made some changes, initially, both rendered but now only the html updates to reflect the changes made" "How do I add a line of best fit to my ggplot figures?" "I used inline r code to refer to the columns and rows within my dataset as such: 'r nrow(campaign' why is my inline r code showing as plain text instead of evaluating?" "if I want to refer to certain column names from my dataset inline during my markdown, what is the syntax to have it display as the greyed out code looking text" "give me a list of all the prompts ive asked you in this thread so far"

Screenshots



🔴 3. Extra or misplaced characters

```R
x <- 5y   # ❌ unexpected symbol
x <- 5*y  # ✅
```

🔴 4. YAML header issues (very common in Quarto)

At the top of your `.qmd` file:

```YAML
---
title: "My Report"
format: html
---
```

Problems happen if:

- You forget quotes around strings with special characters
- You miss a colon (`:`)
- You mix tabs and spaces

🔴 5. Code chunk formatting errors

Make sure chunks look like this:

Ask anything Voice

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🔴 5. Code chunk formatting errors

Make sure chunks look like this:

```Markdown
```{r}
summary(cars)
```
```

Common mistakes:

- Missing backticks
- Typing ```{r} incorrectly (e.g. `{R}` or `{r code}`)
- Putting text inside a code chunk accidentally

🔴 6. Using R syntax in a Python chunk (or vice versa)

If your chunk says:

```Markdown
```{python}
x <- 5   # ❌ R syntax in Python
```
```

That will trigger errors.

🔴 7. Copy-paste issues (hidden characters)

Ask anything Voice

✅ Quick debugging trick

Wrap suspicious lines in comments:

```R
# problematic_line_here
```

Then re-run and narrow it down.

If you want, paste your `.qmd` file or the exact error message here—I can pinpoint the exact line causing the issue.

how do I add a table of content to my quarto doc and number my sections

In a Quarto document, both a table of contents and section numbering are controlled in the YAML header at the top of your `.qmd` file.

✅ Add a table of contents

Just add `toc: true`:

Ask anything

Voice

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✅ Add a table of contents

Just add `toc: true`:

```YAML
---
title: "My Report"
format: html
toc: true
---
```

✅ Number your sections

Add `number-sections: true`:

```YAML
---
title: "My Report"
format: html
toc: true
number-sections: true
---
```

🔧 Optional customization

You can tweak how the table of contents behaves:

Ask anything

Voice

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🔧 Optional customization

You can tweak how the table of contents behaves:

YAML

```yaml
---
title: "My Report"
format: html
toc: true
toc-depth: 3        # how many heading levels to include
toc-location: left  # or "right" or "body"
number-sections: true
---
```

🥾 Make sure your headings are correct

Quarto only builds the TOC from proper Markdown headings:

Markdown

```markdown
# Section 1
## Subsection 1.1
### Sub-subsection 1.1.1
```

⚠️ Common pitfalls

- ❌ Using bold text instead of headings:

Ask anything

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📌 Output format matters

- Works great for: `html`, `pdf`, `docx`
- Some options (like `toc-location`) are mainly for HTML

If you tell me what format you're rendering to (HTML, PDF, Word), I can suggest a cleaner setup—especially for PDFs where TOCs behave a bit differently.

when im trying to render my quarto doc, it updates the html file ive asked it to create but not the pdf file. this is after ive made some changes, initially, both rendered but now only the html updates to reflect the changes made

This usually isn't a Quarto bug—it's almost always a rendering/config issue where the PDF step is either failing silently or not being triggered.

Let's go through the likely causes and fixes.

🔍 1. Check your YAML: are you rendering both formats?

Make sure your header explicitly includes both form

Ask anything

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Make sure your header explicitly includes both formats:

```yaml
---
title: "My Report"
format:
  html: default
  pdf: default
---
```

If it only says `format: html`, then only HTML will update.

🔍 2. Are you actually rendering both outputs?

Depending on how you render:

- **Render button in Positron** → may default to last format used
- Command line:

```bash
quarto render document.qmd
```

→ should render all formats

To force PDF:

```bash
quarto render document.qmd --to pdf
```

Ask anything

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🔴 3. PDF rendering is failing (most common)

What's likely happening:

- HTML renders fine ✅
- PDF fails ❌
- Quarto leaves the *old PDF unchanged*

👉 Check the **render log / terminal output** carefully.

Look for:

- LaTeX errors
- Missing packages
- "Execution halted"

🔧 4. Tiny syntax issues break PDF but not HTML

PDF (via LaTeX) is stricter. Common culprits:

❌ Underscores

Markdown

```
my_variable_name
```

Fix:

Ask anything

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- Missing packages
- "Execution halted"

🔧 4. Tiny syntax issues break PDF but not HTML

PDF (via LaTeX) is stricter. Common culprits:

❌ Underscores

Markdown

```
my_variable_name
```

Fix:

Markdown

```
`my_variable_name`
```

❌ Unescaped symbols

Characters like:

```
% $ # _ { }
```

need escaping in PDF:

Ask anything

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need escaping in PDF:

Markdown

```
50\%
```

❌ Tables / long lines

Wide tables or long inline code can break LaTeX but not HTML.

🔧 5. Try cleaning and re-rendering

Sometimes stale files cause confusion:

Bash

```
quarto clean
quarto render
```

🔧 6. Check if PDF engine is installed

Quarto uses LaTeX (usually via TinyTeX).

Run:

Ask anything

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🔧 6. Check if PDF engine is installed

Quarto uses LaTeX (usually via TinyTeX).

Run:

```bash
quarto check
```

If needed:

```bash
quarto install tinytex
```

🔧 7. Try rendering PDF alone

This helps isolate the issue:

```bash
quarto render document.qmd --to pdf
```

If it fails, you'll see the real error.

🧠 What's probably happening in your case

Ask anything

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🧠 What's probably happening in your case

Since:

> "initially both rendered, now only HTML updates"

Most likely:

👉 A small change introduced a LaTeX-breaking character or syntax
👉 PDF rendering is failing silently
👉 Old PDF is still sitting there unchanged

✅ Quick diagnostic checklist

1. Run:

```bash
quarto render yourfile.qmd --to pdf
```

2. Look for error messages
3. Check for:
   - `_` in text
   - `%` signs
   - weird tables
   - unclosed brackets

Ask anything

Voice

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If you paste:

- your YAML header
- and the error log

I can pinpoint exactly what's breaking the PDF render.

How do I add a line of best fit to my ggplot figures?

To add a line of best fit in **ggplot2**, you use `geom_smooth()` —it automatically computes and overlays a fitted model.

✅ Basic example (linear regression line)

```r
library(ggplot2)

ggplot(mtcars, aes(x = wt, y = mpg)) +
  geom_point() +
  geom_smooth(method = "lm")
```

- `method = "lm"` → fits a **linear model** (straight line)

Ask anything

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🎯 **Remove the confidence interval (shaded area)**

R

```r
geom_smooth(method = "lm", se = FALSE)
```

🎨 **Customize the line**

R

```r
ggplot(mtcars, aes(wt, mpg)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE, color = "red", linewidth = 1)
```

📈 **Non-linear fit (loess)**

R

```r
geom_smooth(method = "loess")
```

- Good for smaller datasets
- Produces a curved “best fit” line

🔢 **Polynomial regression (curved but model-based)**

Ask anything

Voice

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🔢 Polynomial regression (curved but model-based)

```r
geom_smooth(method = "lm", formula = y ~ poly(x, 2))
```

🧠 Common mistakes

- ❌ Forgetting `method = "lm"` → defaults to loess (may look odd on large data)
- ❌ Mapping aesthetics incorrectly:

```r
aes(x = wt, y = mpg)  # must be inside ggplot()
```

- ❌ Using character variables instead of numeric

✅ Multiple groups (separate lines)

```r
ggplot(mtcars, aes(wt, mpg, color = factor(cyl))) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE)
```

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If you want, I can help you:

- add the regression equation onto the plot
- format it nicely for a Quarto report
- or match a specific style your assignment requires 👍

I used inline r code to refer to the columns and rows within my dataset as such: 'r nrow(campaign'
why is my inline r code showing as plain text instead of evaluating?

You're very close—but there's a small syntax issue causing Quarto to treat your inline R as plain text.

❌ What you wrote

Markdown

```
'r nrow(campaign'
```

That won't evaluate because:

- Inline R **must** use backticks, not quotes
- The syntax is strict

Ask anything

Voice

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- Inline R **must** use backticks, not quotes
- The syntax is strict

✅ Correct syntax

Markdown

```
`r nrow(campaign)`
```

So in a sentence:

Markdown

```
The dataset has `r nrow(campaign)` rows.
```

🔑 Key rules for inline R in Quarto

1. **Use backticks** (`), not quotes (' or ")
2. Format must be exactly:

```
`r your_code_here`
```

3. No extra spaces before the `r`

🔴 Common mistakes

Ask anything

Voice

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🔴 Common mistakes

❌ Using quotes

Markdown

```
'r nrow(campaign)'
```

❌ Missing backticks entirely

Markdown

```
r nrow(campaign)
```

❌ Mixing quotes and backticks

Markdown

```
`r nrow(campaign'
```

⚠️ If it's still not working

Check these:

1. Is the document actually rendering?

Inline code only evaluates when you **render**, not just preview raw text.

+ Ask anything Voice

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2. Is the object loaded?

If `campaign` doesn't exist in the document, it won't evaluate.

Make sure earlier you have something like:

```r
campaign <- read.csv("yourfile.csv")
```

---

3. Is it inside a code chunk?

Inline R **does not work inside code chunks**, only in normal text.

❌ This won't work:

```Markdown
```{r}
`r nrow(campaign)`
```
```

🧠 Quick mental model

- Code chunks → big blocks of R
- Inline R → small snippets inside sentences using

Ask anything

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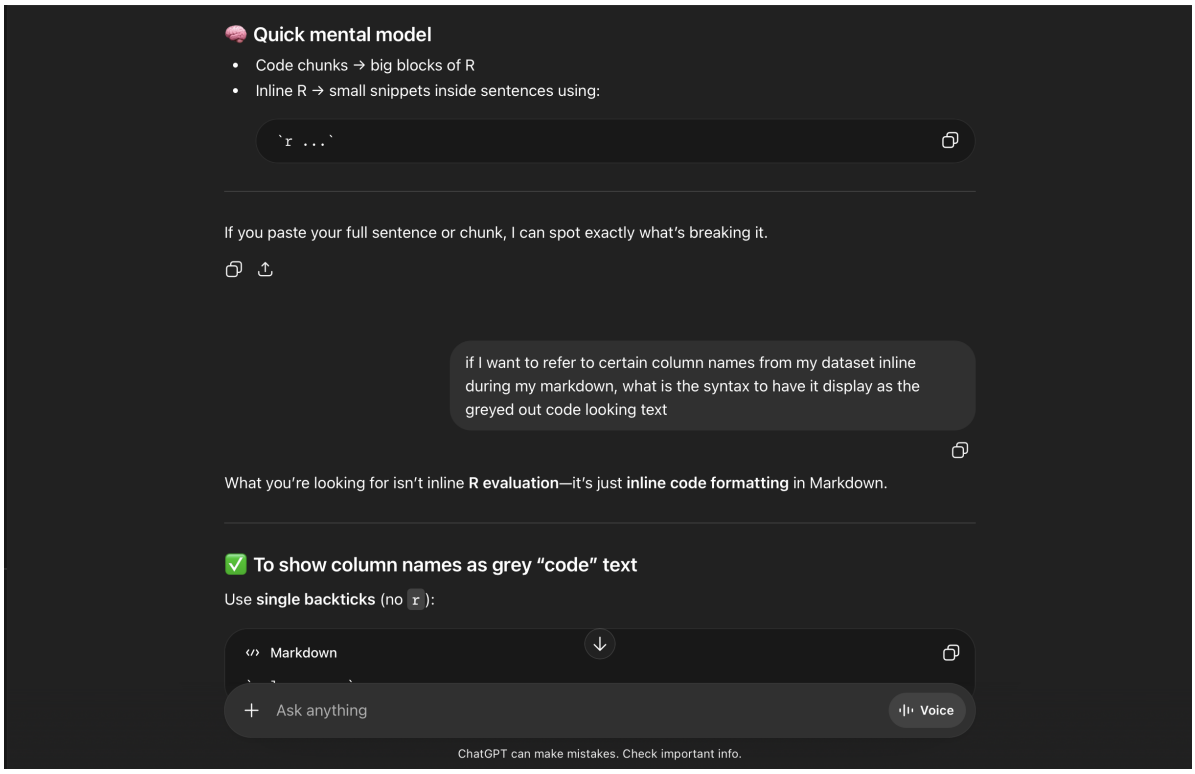
🧠 Quick mental model
Code chunks → big blocks of R
Inline R → small snippets inside sentences using:
`r ...`
If you paste your full sentence or chunk, I can spot exactly what's breaking it.
if I want to refer to certain column names from my dataset inline during my markdown, what is the syntax to have it display as the greyed out code looking text
What you're looking for isn't inline R evaluation—it's just inline code formatting in Markdown.
✅ To show column names as grey "code" text
Use single backticks (no r):
Markdown
Ask anything
Voice
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✅ To show column names as grey "code" text

Use **single backticks** (no `r`):

Markdown

```
`column_name`
```

✏️ Example

Markdown

```
The variable `age` is strongly correlated with `income`.
```

👉 This will render with `age` and `income` in grey code-style formatting.

🔁 Compare the two (important distinction)

1. Inline code formatting (what you want)

Markdown

```
`campaign$age`
```

➡️ Just displays as styled text

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2. Inline R evaluation

Markdown

```
`r nrow(campaign)`
```

➡️ Runs R code and inserts the result

🧠 When to use each

- Use `code formatting` → when referring to column names, variables, file names
- Use `r ...` → when you want computed values (e.g., number of rows, means, etc.)

💡 Pro tip

If your column has spaces or weird characters:

Markdown

```
`campaign$"Customer Age"`
```

If you want, I can show you how to dynamically insert column names (e.g., if they change) using inline R—but for most reports, simple backticks are exactly the right approach.

Ask anything Voice

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If you want, I can show you how to dynamically insert column names (e.g., if they change) using inline R—but for most reports, simple backticks are exactly the right approach.

give me a list of all the prompts ive asked you in this thread so far

Here are all the prompts you've asked in this thread, in order:

1. *"im coding on positron, why does my quarto document show an unexpected symbol error?"*
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Ask anything

Voice

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